9 Fermions and bosons

In this section we set up the statistical mechanics for fermi and bose quantum gases. We uncover the fate of the classical gases for temperatures below $v/\lambda_{\mathrm{T}} \simeq 1$, where the entropy became negative. Bose and fermi gases evade this in very different fashions.

particle / context	Fermions	Bosons
Elementary particles	e^-, ν_e, q ($\mu^\pm$, $\tau^\pm$, ν_μ and ν_τ)	photon, W, Z, gluon, Higgs, graviton
Composites	p, n, ^{3}He, ^{40}K, trions[1]	^{4}He, ^{39}K, H_2, Cooper pair, $\pi^\pm$, π^0
Collective		phonon, spin wave, plasmon
Collections	metal (e^-), nuclei (n,p), white dwarf (e^-), neutron star (n), QGP (q)	superfluid (^{4}He and ^{3}He), superconductor (Cooper pairs), QGP (gluons)

Table 1: Examples of fermions and bosons. (QGP is the quark-gluon plasma.)

What do we mean by ascribing a fermi or bose nature to a composite, such as an atom (eg ^{4}He), or nucleus (eg U^{92+})? Remember that, in a non-relativistic setting, fermions have antisymmetric wave functions and bosons symmetric wave functions – eg for 2 particles, in the same spin state

$$\psi_{\mathrm{F}}(x_1, x_2) = -\psi_{\mathrm{F}}(x_2, x_1) \quad \text{and} \quad \psi_{\mathrm{B}}(x_1, x_2) = +\psi_{\mathrm{B}}(x_2, x_1) \ .$$

The antisymmetry of the fermi wave function implies they cannot reside at the same place, as

$$\psi_{\mathrm{F}}(x_1, x_1) = -\psi_{\mathrm{F}}(x_1, x_1) = 0 \ ,$$

i.e. if a number is equal to minus itself, it must vanish. Including spin (but with no spin-orbit coupling) so the *total* wave-function is (σ_i is the z-component of spin of particle i)

$$\Psi_{\mathrm{F}}(x_1, x_2; \sigma_1, \sigma_2) \ [= \psi_{\mathrm{F}}(x_1, x_2)\chi(\sigma_1, \sigma_2)] = -\Psi_{\mathrm{F}}(x_2, x_1; \sigma_2, \sigma_1) \ ,$$

exchanging *both* coordinates and spin states. Thus one of ψ and χ must be antisymmetric and the other symmetric to ensure an overall antisymmetry.

If we have a *composite* wavefunction – eg for the Hydrogen atom with one proton, X, and one electron, x, then if we consider the wave function for both particles in **two** atoms, (disregarding

[1] A bound state of two electrons and one hole (three fermions), see *Trion sensing of a zero-field composite Fermi liquid*, *Nature*, **635**, 590–595 (2024). *Composite fermions* are also interesting – they are bound states of electrons and flux tubes.

spin for simplicity)

$$\begin{aligned}\psi_{\text{H}}(X_1, x_1; X_2, x_2) &= (-1)(-1)\psi_{\text{H}}(X_2, x_2; X_1, x_1) \\ &= \psi_{\text{H}}(X_2, x_2; X_1, x_1) \ ,\end{aligned}$$

as we have swapped **both** the electrons and the protons, so there are two, cancelling, minus signs. Thus hydrogen atoms are bosons. At low temperatures with atmospheric pressure, hydrogen is a solid composed of H_2 diatomic molecules, so this appears experimentally irrelevant, but see the footnote below.

Thus for composites of an **even** number of fermions (eg ^{4}He), the composites behave as bosons, and for an **odd** number of fermions (eg ^{3}He), the composites are fermions.[2]

Does this matter? Let us examine experiment.

An illustration of the differences is in Fig. (9.1), where the specific heat is shown for both ^{3}He and ^{4}He for $T > 1\,\text{K}$ on the left. There is a phase transition in ^{4}He but not ^{3}He. Given only the nuclei are different this is quite surprising.

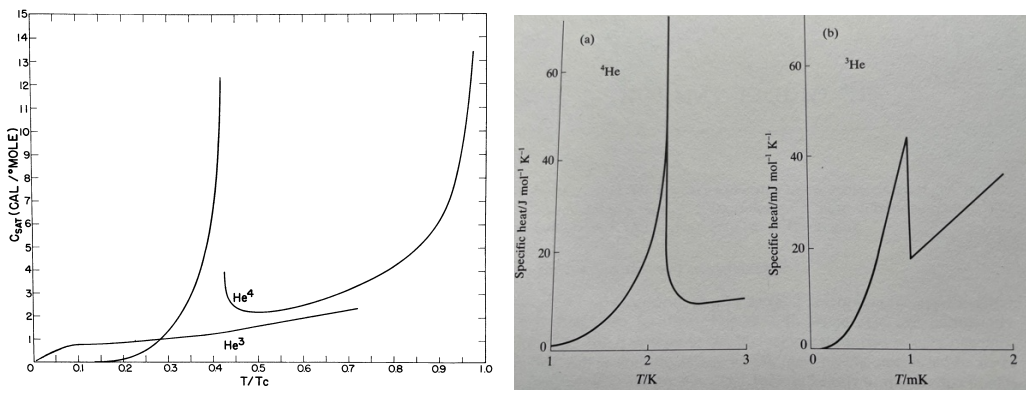

Figure 9.1: The specific heat of the two stable helium isotopes **Left:** the striking difference for the region around 2K between ^{3}He and ^{4}He. **Right:** there is also a phase transition in ^{3}He, but at a temperature a thousand times lower – note difference of horizontal scales. Credit: Betts and Wilks *An introduction to liquid helium*, OUP (1987)

At temperatures of mK ^{3}He also undergoes a phase transition, which resembles that of ^{4}He at 2K, but the magnitude of the anomaly is also about a thousand times smaller – see the difference of vertical scales. We will see the origin of this disparity towards the end of this section. The experimental discovery of the superfluid phases in 1972 (there are at least 3) of

[2]Complications ensue if one squashes the atoms together, so the association of electrons and a particular nucleus becomes unclear. One might make a transition – say in the case of ^{3}He – to two fermi gases (one of electrons, one of helium nuclei). But one might get residual pairing on a scale much bigger than the particle separation, either of the electrons and the nuclei or each of those classes with itself, modelled on the pairing in a superconductor. Speculations about the nature of hydrogen at the high pressures in the centre of Jupiter (it is a molecular solid at atmospheric pressure at $T = 0$) include such scenarios. Eg Cheng, Bingqing ; Mazzola, Guglielmo ; Pickard, Chris J ; Ceriotti, Michele *Nature* (London), **585**, 7824, (2020) and Richardson, C.F ; Ashcroft, N.W *Physical review letters*, **78**, 118 (1997) which contains references to the original proposal.

[3]He required more than a decade of very considerable experimental ingenuity. This is captured in an engaging manner in this short article, from *APS News* October 2021. A new phase was discovered in 2021.

The other important attribute in the distinctions between fermions and bosons is their spin. Relativistic physics, Pauli, W *Physical Review*, **58**, 716 (1940), tells us that fermions have half-integer spin (e.g. electrons have spin 1/2 and ^{40}K have 9/2) and bosons integer spin (e.g. ^{4}He has spin 0 and photons and ^{87}Rb have spin 1).[3]

9.1 Labelling the states

As with waves, for three dimensions we will apply periodic boundary conditions on a cube of linear dimension L. (The following process is readily adapted to fewer or more dimensions.) We further simplify assuming there is no spatially varying potential within the cube. Then the *spatial* part of the wave functions may be taken to be plane waves, in un-normalised form

$$\psi_{\mathbf{k}}(\mathbf{r}) = \exp\left(\mathrm{i}\mathbf{k}\cdot\mathbf{r}\right) \ .$$

The periodic boundary condition along, say, the x-axis is:

$$\begin{aligned} \exp\left(\mathrm{i}\mathbf{k}\cdot\mathbf{r}\right) &= \exp\left(\mathrm{i}\mathbf{k}\cdot[\mathbf{r}+L\hat{\mathbf{x}}]\right) \\ \Rightarrow \quad 1 = \exp\left(\mathrm{i}m_x 2\pi\right) &= \exp\left(\mathrm{i}\mathbf{k}\cdot[L\hat{\mathbf{x}}]\right) \\ &= \exp\left(\mathrm{i}k_x\, L\right) \\ \Rightarrow \quad \frac{2\pi m_x}{L} &= k_x \ , \end{aligned}$$

as before. So using all three directions we see

$$\mathbf{k} = \frac{2\pi}{L}\mathbf{m} \quad \text{where} \quad \mathbf{m} = (m_x, m_y, m_z) \ .$$

We will denote the z-component of the spin variable σ. For electrons $\sigma = \pm\frac{1}{2}$, but the fermionic ^{40}K (potassium) atom has a combined spin (of the sole unpaired electron and the combined nuclear orbital and spin angular momentum) of $F = 9/2$, with $2F+1 = 10$ possible values of σ. ^{181}Ta has a value of the nuclear spin alone of 9.

We will assume the simplifications and define the following terms:

[3]In one and two-dimensions, it is not so clear-cut. There is the possibility of *anyons* which have statistics between fermions and bosons, or worse: the interchanges can have a matrix aspect and the matrices may not commute. This is believed to occur in some of the family of fractional quantum Hall ground states – to go along with their fractionally charged excitations! In three dimensions one needs closed strings to obtain exotic statistics – such as vortex rings in superfluids – they interchange *physically* by the leapfrogging dynamics of concentric smoke rings Niemi, Antti J, *Physical review letters*, **94**, 124502 (2005).

1. We will assume that the particles do not interact. (Sometimes the interactions are important: eg no superconductivity or superfluidity without interactions.)

2. We will initially assume that there is only one type of fermion or boson (e.g. *not* both electrons and protons for example).

3. We will almost always assume there is no potential energy.

4. We will denote the kinetic energy of a single particle in state $\mathbf{k}$ by $\varepsilon_{\mathbf{k}}$. Including a potential dependence on spin (usually, as here, due to a magnetic field) we write

$$\varepsilon_{\mathbf{k}\sigma} = \varepsilon_{\mathbf{k}} - \mu\sigma H \ .$$

5. We define the **occupation number**, $n_{\mathbf{k}\sigma}$ to be the number of particles occupying that state, $\mathbf{k}, \sigma$. If one specifies the whole set of the occupation numbers, $\{n_{\mathbf{k}\sigma}\}$, then the complete quantum state (encompassing all particles) is precisely specified.

6. For fermions $n^{\mathrm{F}}_{\mathbf{k}\sigma} = 0$ or 1, because of the Pauli Exclusion Principle.

7. For bosons, the number can be any integer $n^{\mathrm{B}}_{\mathbf{k}\sigma} \geq 0$.

8. Then the total number of the particles N^{tot} (either fermions or bosons) is for a given set of occupation numbers is

$$N^{\mathrm{tot}}(\{n_{\mathbf{k}\sigma}\}) = \sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma} \ .$$

9. And the total energy, E^{tot},

$$E^{\mathrm{tot}}(\{n_{\mathbf{k}\sigma}\}) = \sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma}\varepsilon_{\mathbf{k}\sigma} \ .$$

9.2 The grand canonical partition function

We may now write down the grand partition function, $\mathcal{Z}$. We use the general procedure, first writing down the number of states:

$$\begin{aligned} W(E,N) &= \sum_{\{n_{\mathbf{k}\sigma}\}} \delta\left(E - E^{\mathrm{tot}}\right) \delta_{N,N^{\mathrm{tot}}} \\ &= \sum_{\{n_{\mathbf{k}\sigma}\}} \delta\left(E - \sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma}\varepsilon_{\mathbf{k}\sigma}\right) \delta_{N,\sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma}} \ . \end{aligned}$$

Here, the first sum is over all possible allocations of the occupation numbers.

Then, as usual, we write (we do not need to initially distinguish bosons and fermions):

$$\begin{aligned}
\mathcal{Z}(T,V,\mu) &= \sum_{N=0}^{\infty}\int \mathrm{d}E\; W(E,N)\exp\left\{-\beta(E-\mu N)\right\} \\
&= \sum_{N=0}^{\infty}\int \mathrm{d}E \sum_{\{n_{\mathbf{k}\sigma}\}} \delta\left(E-\sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma}\varepsilon_{\mathbf{k}\sigma}\right)\delta_{N,\sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma}}\exp\left\{-\beta(E-\mu N)\right\} \\
\text{(Let } \delta\text{'s do their work.)} \quad &= \sum_{\{n_{\mathbf{k}\sigma}\}}\exp\left\{-\beta\left(\sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma}\varepsilon_{\mathbf{k}\sigma}-\mu\sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma}\right)\right\} \\
&= \sum_{\{n_{\mathbf{k}\sigma}\}}\exp\left\{-\beta\left(\sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma}\left[\varepsilon_{\mathbf{k}\sigma}-\mu\right]\right)\right\}
\end{aligned}$$

Now we use the exponential of a sum goes to a product of exponentials result:

$$\begin{aligned}
\mathcal{Z}(T,V,\mu) &= \sum_{\{n_{\mathbf{k}\sigma}\}}\prod_{\mathbf{k}\sigma}\exp\left\{-\beta n_{\mathbf{k}\sigma}\left[\varepsilon_{\mathbf{k}\sigma}-\mu\right]\right\} \\
&= \prod_{\mathbf{k}\sigma}\sum_{n_{\mathbf{k}\sigma}}\exp\left\{-\beta n_{\mathbf{k}\sigma}\left[\varepsilon_{\mathbf{k}\sigma}-\mu\right]\right\} \;,
\end{aligned}$$

where we have distributed the appropriate term in the product to its summation over the corresponding occupation number.

It is at this point we need to distinguish fermions and bosons, as we need to incorporate the upper limit in the sum over the occupation numbers for each state, $\mathbf{k}\sigma$, which is 1 for fermions and ∞ for bosons. Thus, for fermions:

$$\begin{aligned}
\mathcal{Z}^{\mathrm{F}}(T,V,\mu) &= \prod_{\mathbf{k}\sigma}\sum_{n_{\mathbf{k}\sigma}=0}^{1}\exp\left\{-\beta n_{\mathbf{k}\sigma}\left[\varepsilon_{\mathbf{k}\sigma}-\mu\right]\right\} \\
&= \prod_{\mathbf{k}\sigma}\left(1+\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\right) \;.
\end{aligned} \tag{9.1}$$

And for bosons:

$$\begin{aligned}
\mathcal{Z}^{\mathrm{B}}(T,V,\mu) &= \prod_{\mathbf{k}\sigma}\sum_{n_{\mathbf{k}\sigma}=0}^{\infty}\exp\left\{-\beta n_{\mathbf{k}\sigma}\left[\varepsilon_{\mathbf{k}\sigma}-\mu\right]\right\} \\
&= \prod_{\mathbf{k}\sigma}\sum_{n_{\mathbf{k}\sigma}=0}^{\infty}\left\{\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\right\}^{n_{\mathbf{k}\sigma}} \\
\text{(Geometric series.)} \quad &= \prod_{\mathbf{k}\sigma}\frac{1}{1-\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}} \;.
\end{aligned} \tag{9.2}$$

A compact way to summarise Eq. (9.1) and Eq. (9.2) is to write

$$\mathcal{Z}^{\mathrm{F}}_{\mathrm{B}}(T,V,\mu) = \prod_{\mathbf{k}\sigma}\left(1\pm\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\right)^{\pm 1} \;,$$

with the upper sign being the fermi case and the lower sign being the bose case.

9.3 Fermi and bose grand potentials

We now turn to the grand potentials. The expressions below emerge from a consideration of the maximum of the integrand/summand in the combined integral over E and sum over N, in the same manner as we extracted the Helmholtz free energy from the energy integral.

For fermions:

$$\begin{aligned}\Omega^{\mathrm{F}}(T,V,\mu) &= -k_{\mathrm{B}}T\ln\mathcal{Z}^{\mathrm{F}}(T,V,\mu)\\ &= -k_{\mathrm{B}}T\sum_{\mathbf{k}\sigma}\ln\left(1+\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\right)\\ &= \sum_{\mathbf{k}\sigma}\Omega^{\mathrm{F}}_{\mathbf{k}\sigma}(T,V,\mu)\ .\end{aligned} \tag{9.3}$$

For bosons:

$$\begin{aligned}\Omega^{\mathrm{B}}(T,V,\mu) &= -k_{\mathrm{B}}T\ln\mathcal{Z}^{\mathrm{B}}(T,V,\mu)\\ &= -k_{\mathrm{B}}T\sum_{\mathbf{k}\sigma}(-1)\ln\left(1-\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\right)\\ &= k_{\mathrm{B}}T\sum_{\mathbf{k}\sigma}\ln\left(1-\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\right)\\ &= \sum_{\mathbf{k}\sigma}\Omega^{\mathrm{B}}_{\mathbf{k}\sigma}(T,V,\mu)\ .\end{aligned} \tag{9.4}$$

Note the argument of the ln is smaller than unity, so the result for the grand potential will be negative.

We may combine the results as

$$\Omega^{\substack{\mathrm{F}\\ \mathrm{B}}}(T,V,\mu) = \mp k_{\mathrm{B}}T\sum_{\mathbf{k}\sigma}\ln\left(1\pm\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\right)\ ,$$

where the *upper* sign is for fermions and the *lower* sign for bosons.

The advantages of the last equalities above, Eq. (9.3) and Eq. (9.4), is that we can use them to determine the average number of particles in each mode and spin state, $\langle n_{\mathbf{k}\sigma}\rangle$.

Firstly note that

$$\mathrm{d}\Omega(T,V,\mu) = \left(\frac{\partial\Omega}{\partial T}\right)_{V,\mu}\mathrm{d}T + \left(\frac{\partial\Omega}{\partial V}\right)_{\mu,T}\mathrm{d}V + \left(\frac{\partial\Omega}{\partial \mu}\right)_{T,V}\mathrm{d}\mu \tag{9.5}$$

$$\begin{aligned}
\text{(and:)} \qquad \Omega &= F - \mu N \\
\Rightarrow \qquad \mathrm{d}\Omega &= \mathrm{d}F - \mathrm{d}(\mu N) \\
&= -S\,\mathrm{d}T - p\,\mathrm{d}V + \mu\,\mathrm{d}N - \mathrm{d}\mu N - \mu\,\mathrm{d}N \\
&= -S\,\mathrm{d}T - p\,\mathrm{d}V - N\mathrm{d}\mu \qquad (9.6)
\end{aligned}$$

$$\text{(Compare Eq. (9.5), Eq. (9.6).)} \quad \Rightarrow \quad -\frac{\partial\Omega}{\partial\mu} = N\ , \tag{9.7}$$

Then we may use Eq. (9.7) with Eq. (9.3) and Eq. (9.4) for each $\mathbf{k}\sigma$:

$$\begin{aligned}
-\frac{\partial\Omega^{\mathrm{F}}_{\mathrm{B}\,\mathbf{k}\sigma}}{\partial\mu} &= -\mp\frac{\partial}{\partial\mu}k_{\mathrm{B}}T\ln\left(1\pm\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\right) \\
&= \pm k_{\mathrm{B}}T(\pm\beta)\frac{\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}}{1\pm\mathrm{e}^{-\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}} \\
&= \frac{1}{\mathrm{e}^{\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\pm 1} \\
&= \langle n_{\mathbf{k}\sigma}\rangle^{\mathrm{F}}_{\mathrm{B}}\ ,
\end{aligned}$$

where, again, the upper sign is for fermions and the lower sign for bosons. These functions are called the *fermi* and *bose functions* (or fermi and bose *occupation numbers*).

Note: often the angle brackets to explicitly denote thermal averaging are omitted, and they are denoted $n^{\mathrm{F}}(\varepsilon)$ and $n^{\mathrm{B}}(\varepsilon)$, including the sum over spin if there is no magnetic field.

Then for the *total* number of particles we may use the original identity:

$$-\frac{\partial\Omega^{\mathrm{F}}_{\mathrm{B}}}{\partial\mu} = N = \sum_{\mathbf{k}\sigma} n_{\mathbf{k}\sigma} = \sum_{\mathbf{k}\sigma}\frac{1}{\mathrm{e}^{\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}\pm 1}\ .$$

N is a thermodynamic quantity – the total number of particles – so we do not write as a thermal average as the fluctuations are small.

9.3.1 Fermi occupation numbers at low and high temperatures

To interpret the fermi function

$$n_{\mathbf{k}\sigma} = \frac{1}{\mathrm{e}^{\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}+1}\ ,$$

it is simplest to fix $\varepsilon_{\mathbf{k}\sigma}-\mu$ to be positive, negative or zero, and then vary β to determine temperature behaviour.

Behaviour fermi occupation numbers			
Function	$\varepsilon_{\mathbf{k}\sigma} - \mu < 0$	$\varepsilon_{\mathbf{k}\sigma} - \mu = 0$	$\varepsilon_{\mathbf{k}\sigma} - \mu > 0$
$\mathrm{e}^{\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]} \underset{\beta\to\infty}{\sim}$	0	1	∞
$\frac{1}{\mathrm{e}^{\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}+1} \underset{\beta\to\infty}{\sim}$	1	$\frac{1}{2}$	0

The form of the resulting $n_{\mathbf{k}\sigma}(T)$ is sketched in Fig. (9.2). Notice that when $T = 0$,

$$n_{\mathbf{k}\sigma}(T) = \Theta(\mu - \varepsilon_{\mathbf{k}\sigma}) = \begin{cases} 1 & \mu > \varepsilon_{\mathbf{k}\sigma} \\ \frac{1}{2} & \mu = \varepsilon_{\mathbf{k}\sigma} \\ 0 & \mu < \varepsilon_{\mathbf{k}\sigma} \end{cases} ,$$

where $\Theta(x)$ is the Heaviside step-function.

Note the value of n is always 1/2 at the chemical potential (i.e. equal probability of occupation or not) independent of the value of T. At $T = 0$, this is not important for practical purposes!

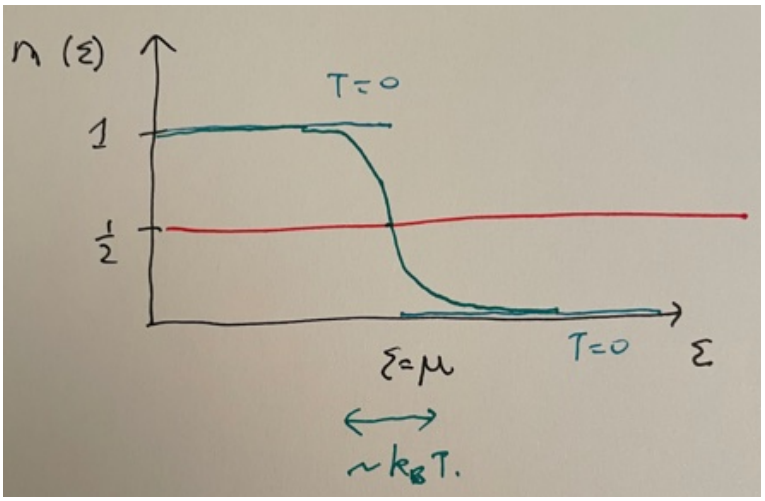

Figure 9.2: The fermi occupation number as a function of energy for zero, infinite and finite temperatures.

The fact that n is energy-independent for $T \to \infty$ may be troubling – it implies there are an infinite number of particles in the system (even before $V \to \infty$). This reflects the fact that the contribution to the grand potential of the entropy per particle is negative and diverges as $T \to \infty$ – as the number of accessible states diverges. Thus each particle added lowers the free energy (by an infinite amount). More concretely, the change in free energy by adding a particle is, in the classical regime

$$\begin{aligned} \Delta F_{\text{particle}} &= \Delta U_{\text{particle}} - T\Delta S_{\text{particle}} \\ &= k_{\mathrm{B}} T \left[\frac{3}{2} - \ln\left(\frac{v}{\lambda_{\mathrm{T}}^3} \right) \right] . \end{aligned}$$

Of course, in reality, T is finite and so there are still energies such that $\varepsilon - \mu \gg k_{\text{B}}T$ where $n \to 0$ and the free energy per particle, while large and negative, is not infinite. But the average number of particles in the system will increase with temperature. This, however, is one of the places where the coulomb interaction alters things for electrons – but not for neutral fermionic atoms (^{3}He etc).

9.3.2 A different view for intermediate temperatures

Let us fix β, i.e. the temperature, and determine how $n_{\mathbf{k}\sigma}(T)$ varies as $\mathbf{k}$ is increased from zero, passing where $\varepsilon_{\mathbf{k}\sigma} = \mu$ and then tends to infinity.

There are three regimes:

Behaviour fermi occupation numbers			
Function	$\beta\left[\varepsilon_{\mathbf{k}\sigma} - \mu\right] \ll 0$	$\beta\left[\varepsilon_{\mathbf{k}\sigma} - \mu\right] \simeq 0$	$\beta\left[\varepsilon_{\mathbf{k}\sigma} - \mu\right] \gg 0$
$\mathrm{e}^{\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}$	0	$1 + \beta\left[\varepsilon_{\mathbf{k}\sigma} - \mu\right]$	∞
$\frac{1}{\mathrm{e}^{\beta[\varepsilon_{\mathbf{k}\sigma}-\mu]}+1}$	1	$\frac{1}{2}\left(1 - \frac{\beta}{2}\left[\varepsilon_{\mathbf{k}\sigma} - \mu\right]\right)$	0

The middle column reveals that over a region of energy $k_{\text{B}}T \simeq |\varepsilon_{\mathbf{k}\sigma} - \mu|$ the fermi occupation number is linear and decreasing with energy. This is represented as the green curve in Fig. (9.2).

This middle region, of extent $k_{\text{B}}T$, determines the thermal effects of the Fermi gas and its response to external stimuli – such as an external magnetic field (see problem sheet 4).

9.4 Complications due to the Coulomb potential

Although we are not taking into account the interaction between the particles, there are some average effects of the Coulomb interaction in the case of electrons (as against neutral fermions such as neutrons, neutrinos, ^{3}He and ^{40}K) which cannot be ignored.

If one adds extra electrons to a three-dimensional conductor, they reside on the surface (as

discussed in Y1 electromagnetism), and one cannot change the *total* number by a finite fraction. So in reality one is always working with a fixed number of particles determined by the compensating positive nuclear charges, with any effect due to voltages of leads etc restricted to $\Delta n \propto L^2/L^3$, reflecting the ratio of surface area to volume.

In **two dimensions**, there is freedom to change the charge density. This is very important in designing and constructing semiconductor structures with elements such as field-effect transistors. There, the "sample" may be thought of as one plate of a parallel plate capacitor, the other being the *gate*. By changing the voltage of the gate (and hence the charge on it), the charge in the sample may be altered (by orders of magnitude). A less flexible, but more delicate, method is *modulation doping* which builds a "capacitor" into a layered semiconductor sandwich with a fixed charge on both "plates".

What about the protons in nuclei? The strong interaction and kinetic energy associated with confinement dominate over the electrostatics for light nuclei – until the issue of deformation and eventual fission occurs in heavy nuclei, eg uranium, caused by electrostatics.

9.5 The continuum limit

We will now exchange the $\mathbf{k}$-space sums in the grand potential for integrals over energy of the particles, which will provide some simplification and suggest an approximation scheme at low temperatures.

As usual we write, assuming $2S+1$ spin states,

$$\sum_{\mathbf{k}\sigma} = (2S+1)\left(\frac{L}{2\pi}\right)^d S_d \int_0^\infty \mathrm{d}k\ k^{d-1}\ ,$$

where S_d is the area of the unit hypersphere in d-dimensions, i.e.

$$S_1 = 2\ ;\ S_2 = 2\pi\ ;\ S_3 = 4\pi \quad \text{etc}\ ,$$

which may be regarded as the result of the "angular" integrals, in three dimensions due to θ_k and ϕ_k.

We assume the kinetic energy has the simple form, independent of σ:

$$\varepsilon_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m^*}\ ,$$

allowing for an *effective mass*, m^*, in a condensed matter system[4]. Then, changing variables

[4] This can differ from the "bare" mass due to the effect of the periodic potential the electrons move through, or the interactions between the electrons, or with phonons, spin waves etc.

from k to ε:

$$
\begin{aligned}
k &= \frac{\sqrt{2m^*\varepsilon}}{\hbar} \\
\Rightarrow \quad \mathrm{d}k &= \frac{\sqrt{2m^*}}{\hbar}\frac{1}{\sqrt{\varepsilon}}\frac{1}{2}\,\mathrm{d}\varepsilon \\
\Rightarrow \quad k^{d-1}\mathrm{d}k &= \frac{1}{2}\left(\frac{2m^*}{\hbar^2}\right)^{d/2}\varepsilon^{(d-2)/2}\,\mathrm{d}\varepsilon \ .
\end{aligned}
$$

So we may write the expression for the grand potential as:

$$
\begin{aligned}
\Omega^{\mathrm{F}}_{\mathrm{B}}(T,V,\mu) &= \mp k_{\mathrm{B}}T(2S+1)\left(\frac{2m^*L^2}{h^2}\right)^{d/2}\frac{S_d}{2}\int_0^\infty \mathrm{d}\varepsilon\ \varepsilon^{(d-2)/2}\ln\left(1\pm \mathrm{e}^{-\beta[\varepsilon-\mu]}\right) \\
&= \mp k_{\mathrm{B}}T\int_0^\infty \mathrm{d}\varepsilon\ g(\varepsilon)\ln\left(1\pm \mathrm{e}^{-\beta[\varepsilon-\mu]}\right) \ ,
\end{aligned}
$$

where we have defined the *density of states*, $g(\varepsilon)$:

$$
g(\varepsilon) = (2S+1)\,\frac{S_d}{2}\left(\frac{2m^*L^2}{h^2}\right)^{d/2}\varepsilon^{(d-2)/2} \ . \tag{9.8}
$$

We may use the same changes of variable to deduce expressions for the total particle number, N, and the internal energy, U:

$$
N = \sum_{\mathbf{k},\sigma} n_{\mathbf{k}\sigma} \qquad = \int_0^\infty \mathrm{d}\varepsilon\ \frac{g(\varepsilon)}{\mathrm{e}^{\beta[\varepsilon-\mu]}+1} \tag{9.9}
$$

$$
U = \sum_{\mathbf{k},\sigma} \varepsilon_{\mathbf{k}\sigma} n_{\mathbf{k}\sigma} \qquad = \int_0^\infty \mathrm{d}\varepsilon\ \frac{\varepsilon\, g(\varepsilon)}{\mathrm{e}^{\beta[\varepsilon-\mu]}+1} \ . \tag{9.10}
$$

These equations define the thermodynamics of the fermi gas. We may, for example (in principle), solve Eq. (9.9) for $\mu(N)$ and substitute into Eq. (9.10) to obtain $U(T,V,N)$.

However the integrals may not be performed in terms of elementary functions and analytically one may only deduce their behaviour at $T=0$ (easy), high temperatures (relatively easy, yielding classical results) and low temperatures (rather harder). At low temperatures the standard analysis requires the *Sommerfeld expansion* (see, for example, p 363-365 of e-book *Concepts in thermal physics*, Stephen J. Blundell and Katherine M. Blundell. 2nd edition, OUP (2010)). It revolves around noting that the derivative of $\left(\mathrm{e}^{\beta[\varepsilon-\mu]}\right)^{-1}$ is sharply peaked at $\varepsilon=\mu$ to simplify the integrals by Taylor-expanding the rest of the integrand around $\varepsilon=\mu$. The calculation is made harder by the coupled nature of calculating μ to find the consistent power series in T for both N (or μ) and U.

9.6 Fermi gases at zero and low temperatures

Firstly let us evaluate Eq. (9.9) and Eq. (9.10) at $T = 0$, allowing us to get a feeling for the scale of internal energy contained in the fermi gas. This is a completely quantum result – remember the domination of entropy when we considered the free energy of a classical ideal gas. It provides the correct picture of the low temperature region where the classical entropy became negative. We will also see the scale involved in a variety of real systems.

At $T = 0$ all states up to $\varepsilon = \mu(T = 0) = \varepsilon_{\text{F}}$ are occupied. The term "fermi energy" is used most accurately as the limit of μ as $T \to 0$. Then in k-space, all the states with kinetic energy, $\hbar^2 k^2/(2m^*) < \mu = \varepsilon_{\text{F}}$ are occupied those with higher energies are all empty.

The boundary between the occupied and unoccupied states is at

$$\begin{aligned} \varepsilon_{\text{F}} &= \frac{\hbar^2}{2m}\mathbf{k}^2 = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right) \\ \Rightarrow \quad \frac{2m\varepsilon_{\text{F}}}{\hbar^2} &= k_x^2 + k_y^2 + k_z^2 \ , \end{aligned}$$

i.e. a sphere in k-space of radius $\sqrt{2m\varepsilon_{\text{F}}}/\hbar$. The states *inside* the sphere are occupied, and those *outside* are unoccupied.

The spherical region of occupied states with radius k_{F} is termed, poetically, the *fermi sea*. The surface of the sphere is k-space is called the *fermi surface*. The vicinity of the fermi surface is the arena for most fermi physics in a metal before it melts. Apart from the lefthand-most column of the periodic table (Li, Na, K etc) the fermi surfaces of real metals are more complex than a sphere, for example that of Cu in Fig. (9.3), due to the periodic potential in which the electrons move. The periodicity is because of the Brillouin Zones, and the "necks"

The fermi surface of Cu is actually quite simple still; the fermi surfaces of the other elements are in general much more complicated – see the wonderful "periodic table of fermi surfaces". Note that the left hand column are unaffected by the periodic potential (potassium, sodium .. except caesium) and the noble metals (of which copper is an example) are the next least affected.

Firstly we will deduce the relation between the fermi energy, ε_{F} and the number of fermions.

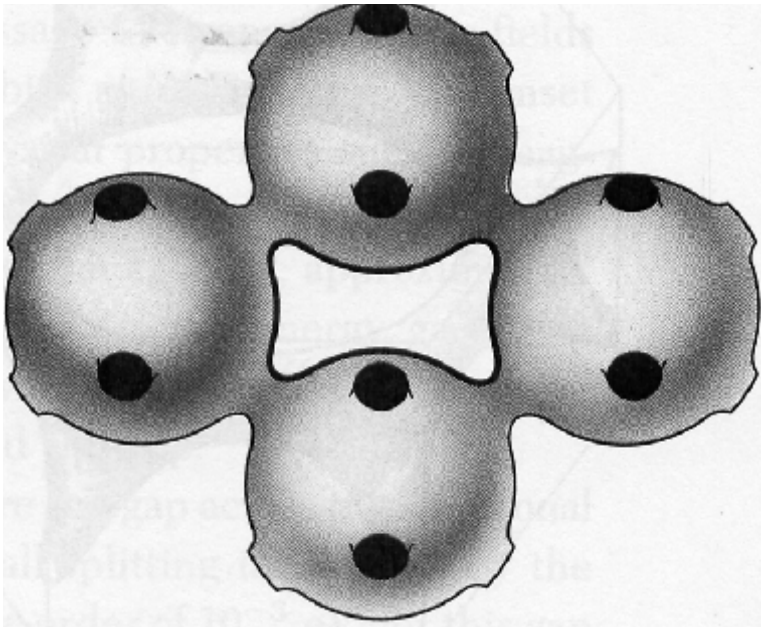

Figure 9.3: The fermi surface of Cu. Credit: M C Chang, Uni. Taiwan

Using Eq. (9.8) we see:

$$
\begin{aligned}
N &= \int_0^{\varepsilon_{\mathrm{F}}} \mathrm{d}\varepsilon\; g(\varepsilon) \\
&= (2S+1)\frac{S_d}{2}\left(\frac{2m^* L^2}{h^2}\right)^{d/2}\left[\frac{\varepsilon^{(d-2)/2+1}}{(d-2)/2+1}\right]_0^{\varepsilon_{\mathrm{F}}} \\
\Rightarrow \quad N &= (2S+1)\frac{S_d}{2}\left(\frac{2m^* L^2}{h^2}\right)^{d/2}\frac{2}{d}\,\varepsilon_{\mathrm{F}}^{d/2} \\
\left(\text{Let } n = \frac{N}{V} = \frac{N}{L^d}\right) \quad \Rightarrow \quad \varepsilon_{\mathrm{F}} &= \left(\frac{(2\pi)^d\, d}{S_d(2S+1)}\right)^{2/d}\frac{\hbar^2 n^{2/d}}{2m^*}\,, \qquad (9.11)
\end{aligned}
$$

noting the effect of the 2π's in restoring $h \to \hbar$. Then for the three-dimensional result:

$$
\varepsilon_{\mathrm{F}} = \left(\frac{(2\pi)^3\, 3}{4\pi\,(2S+1)}\right)^{2/3}\frac{\hbar^2 n^{2/3}}{2m^*} = \left(\frac{6\pi^2}{(2S+1)}\right)^{2/3}\frac{\hbar^2 n^{2/3}}{2m^*} \qquad (9.12)
$$

We could have derived – or rather guessed – the form by dimensional analysis. The only length scale provided by the problem is L, but $\hbar^2/2mL^2$ is the energy of the ground state which is not helpful for fermions. So one might guess that the number density, $N/L^3 = n$, is right combination, and then $\hbar^2/2m$ part is forced.

Let us turn to U. The integral is of the same form, and indeed we can use Eq. (9.11) to

simplify considerably.

$$
\begin{aligned}
U &= \int_0^{\varepsilon_{\mathrm{F}}} \mathrm{d}\varepsilon \; g(\varepsilon) \; \varepsilon \\
&= (2S+1)\frac{S_d}{2}\left(\frac{2m^* L^2}{h^2}\right)^{d/2} \left[\frac{\varepsilon^{(d/2)+1}}{(d/2)+1}\right]_0^{\varepsilon_{\mathrm{F}}} \\
\text{(Use Eq. (9.11).)} \quad &= N\frac{d/2}{(d+2)/2}\varepsilon_{\mathrm{F}} \; . \qquad (9.13)
\end{aligned}
$$

For three dimensions we see:

$$U = \frac{3}{5}\varepsilon_{\mathrm{F}} N \; .$$

This is to be expected – the average energy of the particles is less than, but of the same order as ε_{F}.

The fermi energies for some real systems are in the following table. A convenient unit is electron-volts, $1\,\mathrm{eV} \simeq 1.6 \times 10^{-19}\,\mathrm{J}$, where the ionisation energy of the hydrogen atom is roughly 13 eV and the rest mass energy of a proton is a bit less than 1 GeV.

Experimental Fermi energies						
System	Electrons in a metal	^{3}He liquid	^{40}K ultracold gas	Protons in a nucleus	Electrons in a white dwarf	Neutrons in a neutron star
Fermi energy	10 eV	0.1 meV	0.1 μeV	40 MeV	0.3 MeV	100 MeV

A striking aspect of the fermi gas at zero temperature (compared to the classical gas) is that the pressure remains finite at $T = 0$ because the fermions still have momentum due to the Pauli principle, so can still exert a pressure due to the reversal of their momentum at boundaries.

$$
\begin{aligned}
p &= -\frac{\partial U}{\partial V} && (9.14)\\
&= -\frac{d}{(d+2)} N \frac{\partial \varepsilon_{\mathrm{F}}}{\partial V} \\
&= -\frac{d}{(d+2)} N \left(\frac{(2\pi)^d\, d}{S_d(2S+1)} \right)^{2/d} \frac{\partial}{\partial V} \frac{\hbar^2 n^{2/d}}{2m^*} \\
&= -\frac{d}{(d+2)} N \left(\frac{(2\pi)^d\, d}{S_d(2S+1)} \right)^{2/d} \frac{\hbar^2 n^{2/d}}{2m^*} \left(-\frac{2}{d} \right) \frac{1}{V} \\
&= \frac{2}{d+2}\, n\, \varepsilon_{\mathrm{F}} && (9.15)\\
\text{(Use Eq. (9.13).)} \quad \Rightarrow \quad p &= \frac{2}{d} \frac{U}{V} \,. && (9.16)
\end{aligned}
$$

The pressure itself is not directly measurable, but the bulk modulus, B, is:

$$
\begin{aligned}
B &= -V \frac{\partial p}{\partial V} \\
\text{(Use Eq. (9.16).)} \quad &= -V \frac{2}{d} \left(\frac{1}{V} \frac{\partial U}{\partial V} - \frac{U}{V^2} \right) \\
\text{(Use Eq. (9.14).)} \quad &= \frac{2}{d} p + \underbrace{\frac{2}{d} \frac{U}{V}}_{=p} \\
&= \frac{d+2}{d}\, p \\
\text{(Use Eq. (9.15).)} \quad &= \frac{2}{d}\, n \varepsilon_{\mathrm{F}} \,.
\end{aligned}
$$

The bulk moduli for the alkali metals (the elements in the first column which have spherical fermi surfaces) and the noble metals (next most complicated) are shown in Fig. (9.4), showing the heavier alkali metals agree well with the estimate above.

Table 30.1 Properties of selected metals

n (10^{28} m^{-3})	E_F (eV)	$\frac{2}{3} n E_{\mathrm{F}} (10^9 \mathrm{Nm}^{-2})$	B(10^9Nm^{-2})	
Li	4.70	4.74	23.8	11.1
Na	2.65	3.24	9.2	6.3
K	1.40	2.12	3.2	3.1
Cu	8.47	7.00	63.3	137.8
Ag	5.86	5.49	34.3	103.6

Figure 9.4: The bulk moduli for the alkali and noble metals predicted by ideal fermi gas and experimental results. Credit: Blundell and Blundell *Concepts in thermal physics*, OUP (2010).

For three dimensions we have:

$$
U = \frac{3}{5} n \varepsilon_{\mathrm{F}} \quad \text{and} \quad p = \frac{2}{3} \frac{U}{V} = \frac{2}{5} n \varepsilon_{\mathrm{F}} \quad \text{and} \quad B = \frac{10}{9} \frac{U}{V} = \frac{5}{3} p = \frac{2}{3} n \varepsilon_{\mathrm{F}} \,.
$$

Let us turn to the low, but not zero, temperature limit. We will choose the specific heat, because the specific heat at low temperatures conveys information about effects beyond the ideal gas (see below). Deriving the precise form is a bit involved (primarily because of the interplay between μ and U mentioned above), so an approximate method will be used (for the more careful derivation, see, for example, p 363-365 of e-book *Concepts in thermal physics*, Stephen J. Blundell and Katherine M. Blundell. 2nd edition, OUP (2010)).

Most of the electrons at low temperatures are in states whose occupancy is always unity. However there is a shell of electronic states within $k_{\rm B}T$ of the chemical potential where the occupancy is less than unity, and hence contribute to the entropy and the specific heat. A simple approximation would be that the number of electrons involved in thermal processes would be $g(\varepsilon = \mu)k_{\rm B}T$, where $g(\varepsilon)$ is the density of states per unit energy. If we ascribe a specific heat of $3/2R$ to each such electron, as if they were classical particles, we would obtain:

$$C_V \simeq g(\varepsilon = \mu)k_{\rm B}T\frac{3}{2}k_{\rm B} \ . \tag{9.17}$$

If we now use Eq. (9.8), let the effective mass be m^*, and assume that at low enough temperatures, $\mu \simeq \varepsilon_{\rm F}$, then:

$$\begin{aligned}
g(\mu) \simeq g(\varepsilon_{\rm F}) &= \frac{1}{2\pi^2}\left(\frac{2m^*L^2}{\hbar^2}\right)^{3/2}\varepsilon_{\rm F}^{1/2} \\
&= \frac{N}{2\pi^2}\left(\frac{2m^*}{\hbar^2 n^{2/3}}\right)^{3/2}\varepsilon_{\rm F}^{1/2} \\
\left(\text{Note, using Eq. (9.11)} \quad \varepsilon_{\rm F}^{3/2} = 3\pi^2\left(\frac{\hbar^2 n^{2/3}}{2m^*}\right)^{3/2}\right) \\
&= \frac{3N}{2}\frac{1}{\varepsilon_{\rm F}^{3/2}}\varepsilon_{\rm F}^{1/2} \\
\Rightarrow \quad g(\mu) &= \frac{3N}{2}\frac{1}{\varepsilon_{\rm F}}
\end{aligned} \tag{9.18}$$

We now substitute Eq. (9.18) into Eq. (9.17) to obtain:

$$\begin{aligned}
C_V &= g(\varepsilon = \mu)k_{\rm B}T\frac{3}{2}k_{\rm B} \\
&= \left(\frac{3N}{2}k_{\rm B}\right)\frac{3}{2}\frac{k_{\rm B}T}{\varepsilon_{\rm F}} \\
&= \gamma T \ ,
\end{aligned} \tag{9.19}$$

defining the quantity γ, which is proportional to the effective mass, m^*. We will return to this presently. The exact term at order T^2 is:

$$C_V^{\text{Sommerfeld}} = \frac{3N}{2}k_{\rm B}\left(\frac{\pi^2}{3}\frac{k_{\rm B}T}{\varepsilon_{\rm F}}\right) + O(T^3) \ .$$

An experimental fact is explained by the suppression of the electronic contribution to the specific heat by $k_{\rm B}T/\varepsilon_{\rm F}$ compared to a classical gas. Before this result was derived by Sommerfeld it was expected that metals would have a much larger specific heat than insulators

because of the extra electron per atom (roughly). But the specific heats are the same order of magnitude, dominated by the phonon contribution which is the same order of magnitude, except at low temperatures. This is because of the suppression of the electronic contribution – until a temperature of 10^4 K!

Since $\gamma \propto m^*$ (as $\varepsilon_{\rm F} \propto 1/m^*$), the specific heat is a method to measure the effect on m^* of the periodic potential or interactions between the particles or some other "quasi-particles" such as phonons or spin waves.

To see how interactions amongst the particles can dramatically change the specific heat see Fig. (9.5). Remembering that $C_V \propto m^*$, we note that the scale of the curves for Au and YbBiPt differ by almost three orders of magnitude, indicating that for YbBiPt $m^* \sim 1000\, m_0$, where m_0 is the "bare" mass of an electron in free space (for Au $m^* \simeq 1.1 m_0$). The evidence that this is due to interactions amongst the particles is that a relatively small magnetic field (~ 0.4 T) restores the effective mass to the order of m_0, see Mun, E et al *Phys. Rev. B* **92**, 085135 (2015). The peak at $T = 0.4$ K reflects a transition to antiferromagnetic spin order.

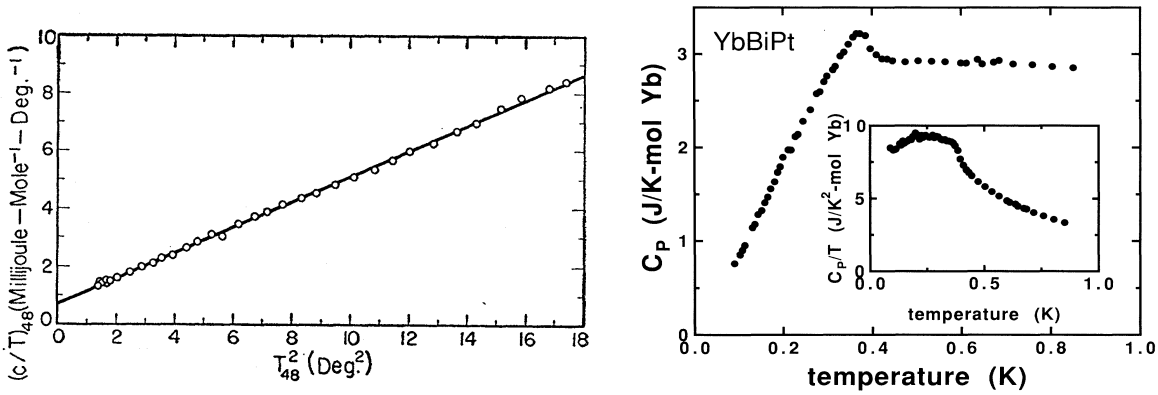

Figure 9.5: Left: The specific heat of Au. **Right:** The specific heat of YbBiPt. Note the vertical scale in terms of energy is a thousand times larger on the right (and the scale in linear in T on the right and quadratic on the left). Credits: Corak, William S et al *Physical Review*, **98**, 1699 (1955) and Fisk, Z et al, *Phys. Rev. Lett.*, **67**, 3310 (1991) respectively.

In ^{3}He, an indication of more subtle effects is that the specific heat for temperatures $T \lesssim 100$ mK is

$$C_V \simeq \gamma T + \Gamma T^3 \ln(T/\Theta_{\rm c}) \ ,$$

where γ, Γ and $\Theta_{\rm c} \simeq 0.5$ K are positive constants. Note the second term is *negative* in this region of temperature. Also it is not obtainable as a power series in T – unlike the corrections to $C_V \sim \gamma T$ for the ideal fermi gas.

The second term is associated with ferromagnetic fluctuations in the (*nuclear*) spins which are associated with the binding of the ^{3}He atoms into Cooper pairs in the superfluid state. See Fig. (9.8), where the variables along the axes have been chosen to reveal the logarithmic term. The experiments by Greywall were a tour de force in very accurate measurements at such low temperatures.

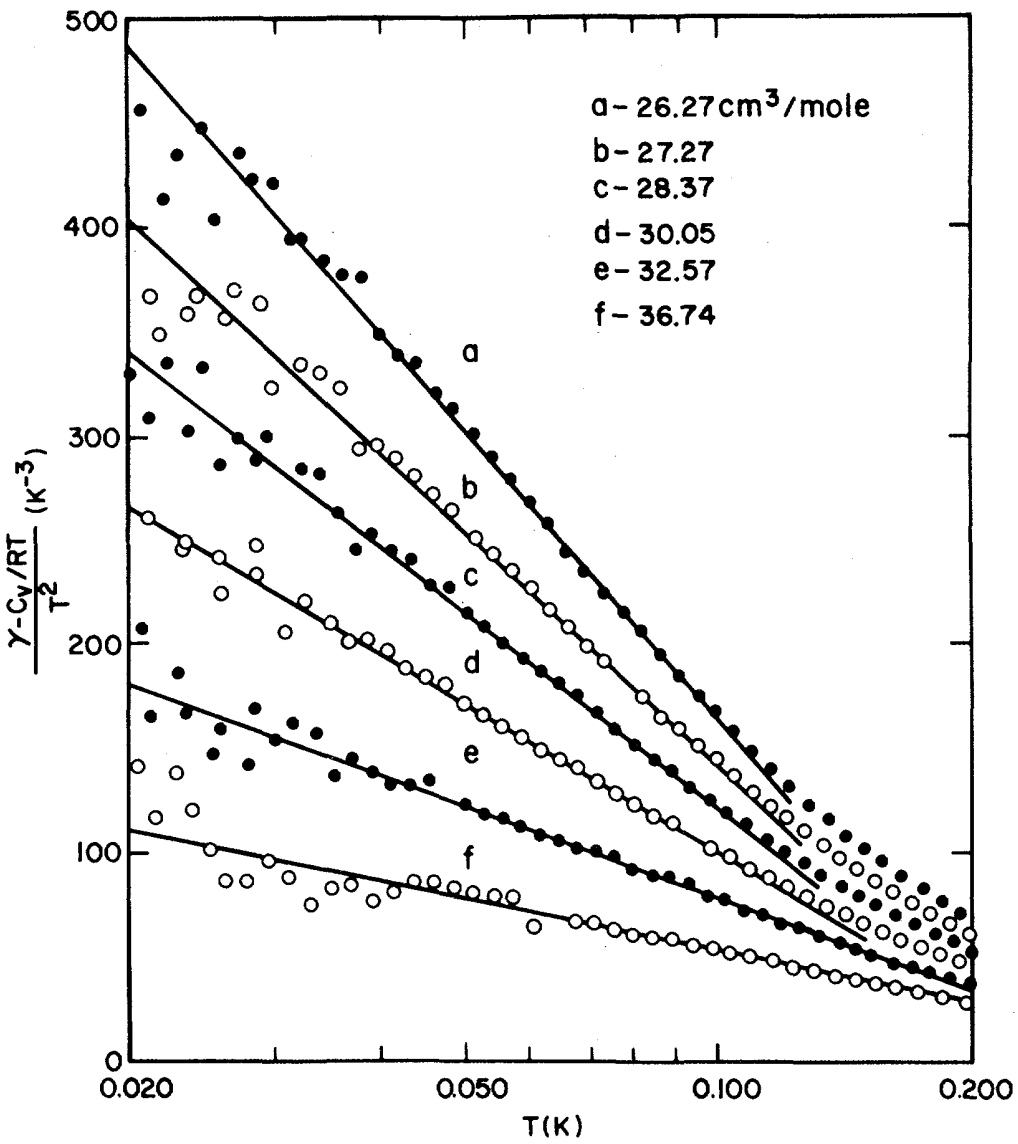

Figure 9.6: The low temperature specific heat for ^{3}He, with axes chosen to extract the logarithmic term as a straight line (note the log scale on the horizontal axis). Credit: Greywall, Dennis S, *Physical Review. B*, **27**, 2747 (1983).

A final oddity about ^{3}He is that there is a minimum size of droplet – around 30-40 atoms are required for a bound state (Sola, E, *Physical Review. B*, **73**, 092515, (2006)). On the other hand, ^{4}He even has a two atom bound state (Grisenti, RE et al *Physical Review Letters*, **85**, 2284 (2000)), He_2, which has been studied experimentally. It is very large (~ 52 Å, compared to the H_2 molecule at 1.2 Å) due to the binding being caused by the van der Waals interaction and not "chemistry". The smallest ^{3}He clusters studied experimentally have around 1000 atoms, Harms, Jan et al *Physical Review. B*, **63**, 184513 (2001).

9.7 Interlude on the chemical potential for both classical and quantum gases

Let us remember that the chemical potential, $\mu(T, V, N)$, of the classical ideal gas is

$$\mu = -k_{\mathrm{B}} T \ln\left(\frac{v}{\lambda_{\mathrm{T}}^3}\right) , \tag{9.20}$$

where the volume per particle is v and the thermal de Broglie wavelength, λ_{T}, are:

$$v = \frac{V}{N} \quad \text{and} \quad \lambda_{\mathrm{T}} = \sqrt{\frac{\mathsf{h}^2}{2\pi m k_{\mathrm{B}} T}} .$$

This expression should not be trusted when $v/\lambda_{\mathrm{T}}^3 \leq 1$, where the entropy is negative, at sufficiently low temperatures – as discussed in section 5.

We have implicitly seen that for fermions, μ will be classical when $k_{\mathrm{B}} T \gg \varepsilon_{\mathrm{F}}$, and there is no remanent of the fermi sea. μ grows as T decreases crossing $\mu = 0$ approximately where the classical entropy becomes problematic. As $T \to 0$, the chemical potential obeys (Sommerfeld expansion)

$$\mu(T, V, N) \simeq \varepsilon_{\mathrm{F}} \left[1 - \frac{\pi^2}{12}\left(\frac{k_{\mathrm{B}} T}{\varepsilon_{\mathrm{F}}}\right)^2 - \cdots\right] ,$$

i.e. approaches ε_{F} from below. The balance of the contributions to μ (remember it is the change in free energy to add on particle) changes as μ becomes positive from predominantly entropy to predominantly internal energy.

In the case of bosons, we will find a different resolution below. See Fig. (9.7), where the contrasting behaviour of $\mu(T)$ for the classical, fermionic and bosonic ideal gases are depicted.

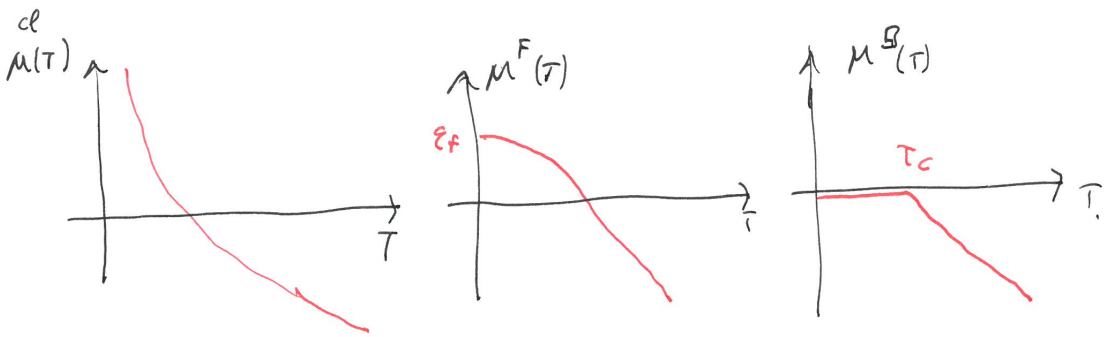

Figure 9.7: The chemical potential, $\mu(T)$, as a function of temperature for: the classical ideal gas; a fermi ideal gas; and an ideal bose gas.

9.8 Bosons and Bose-Einstein condensation

In this section we will investigate the behaviour of Bose particles as $T = 0$ is approached, and find a dramatic change at a temperature determined by the product $n\lambda_{\mathrm{T}}^3$, i.e. the number of particles within a thermal de Broglie volume.

Let us start from the expression for the average total particle number, N,

$$N = \sum_{\mathbf{k}\sigma} n^{\mathrm{B}}_{\mathbf{k}\sigma} = \sum_{\mathbf{k}\sigma} \frac{1}{\mathrm{e}^{\beta(\varepsilon_{\mathbf{k}}-\mu)} - 1} \ .$$

Notice the energy of the lowest state with $\mathbf{k} = 0$, $\varepsilon_0 = 0$, must obey $\varepsilon_0 - \mu > 0$, i.e. $\mu \leq 0$ as otherwise the $n^{\mathrm{B}}_{\mathbf{k}\sigma}$ might be negative which is unphysical.

Let us now take the continuum limit, and we will encounter a problem. The continuum limit of the equation for N is:

$$N = \int_0^\infty \mathrm{d}\varepsilon \ g(\varepsilon) n^{\mathrm{B}}(\varepsilon) = \int_0^\infty \mathrm{d}\varepsilon \ g(\varepsilon) \frac{1}{\mathrm{e}^{\beta(\varepsilon-\mu)} - 1} \ , \tag{9.21}$$

where the density of states, $g(\varepsilon)$, is the same as a fermi gas (apart from the spin, S, being an integer) – see Eq. (9.8)

$$g(\varepsilon) = \underbrace{(2S+1)2\pi \frac{(2m)^{3/2}}{\mathsf{h}^3}}_{=C} V \varepsilon^{1/2} = C\, V \varepsilon^{1/2} \ ,$$

where C is a constant.

Since $n^{\mathrm{B}}(\varepsilon)$ increases as $\mu \leq 0$ increases, the maximum value of N occurs at $\mu = 0$ (letting $x = \beta\varepsilon$):

$$\begin{aligned} N_{\mathsf{max}} &= CV \int_0^\infty \mathrm{d}\varepsilon \frac{\varepsilon^{1/2}}{\mathrm{e}^{\beta(\varepsilon-\mu)} - 1}\bigg|_{\mu=0} \\ &= \frac{CV}{\beta^{3/2}} \int_0^\infty \frac{x^{1/2}\,\mathrm{d}x}{\mathrm{e}^x - 1} \\ &= \frac{CV}{\beta^{3/2}} \Gamma\left(\tfrac{3}{2}\right) \zeta\left(\tfrac{3}{2}\right) \\ &\simeq 2.315\, CV (k_{\mathrm{B}}T)^{3/2} \ . \end{aligned} \tag{9.22}$$

Where the Gamma and Riemann Zeta functions are defined (see also the table Baierlein, *Thermal Physics*, CUP (1999), Appendix A, p422):

$$\Gamma(z) = \int_0^\infty \mathrm{d}t \ \mathrm{e}^{-t} t^{z-1} \ , \qquad \zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s}$$

$$\text{with} \quad \Gamma\left(\tfrac{3}{2}\right) = \tfrac{1}{2}\sqrt{\pi} \ , \qquad \zeta\left(\tfrac{3}{2}\right) \simeq 2.612 \ .$$

But this is nonsense – there is no limit to the number of bosons one may have in the system. Note that $N_{\text{max}} \to 0$ as $T \to 0$.

We need to isolate the lowest state, which (with periodic boundary conditions) occurs at $\mathbf{k} = 0$, implying $\varepsilon_0 = 0$. This state, residing at the lower limit of integration gets lost because of the *average* factor of $\varepsilon^{1/2}$ in the integrand. Or, to put it another way, smoothing out the delta functions at each allowed value of the energy fails in this particular case[5], so the N in Eq. (9.21) is actually the number of particles in the excited states, $N_{\neq 0}$. Then, letting N_0 be the occupation of the ground state:

$$\begin{aligned} N &= N_0 + N_{\neq 0} \\ &= N_0 + \frac{CV}{\beta^{3/2}} \int_0^\infty \frac{x^{1/2}\,\mathrm{d}x}{\mathrm{e}^x - 1} \ . \end{aligned} \tag{9.23}$$

The interpretation of the above statements is maybe stated if one fixes V and T. Then there is a maximum number of bosons, N_{max}, occupying the excited states which is proportional to V (and depending on T). If bosons are added beyond that, the increase in total number, N, is accommodated by increased occupation of the ground state: N_0 will grow.

Consider fixing N and lowering the temperature. When the chemical potential, $\mu(T)$, vanishes as the temperature is lowered, at $T = T_{\text{c}}$, we find an equation which determines T_{c} in terms of N (which equals N_{max} at that T), V and constants:

$$\begin{aligned} N = N_{\text{max}}(T_{\text{c}}) &= \frac{CV}{\beta_{\text{c}}^{3/2}} \int_0^\infty \frac{x^{1/2}\,\mathrm{d}x}{\mathrm{e}^x - 1} \\ &= CV\,(k_{\text{B}}T_{\text{c}})^{3/2}\,\Gamma\left(\tfrac{3}{2}\right)\zeta\left(\tfrac{3}{2}\right) \end{aligned} \tag{9.24}$$

$$\begin{aligned} \Rightarrow \quad N &= (2S+1)\frac{2\Gamma\left(\frac{3}{2}\right)\zeta\left(\frac{3}{2}\right)}{\sqrt{\pi}} V \frac{(2\pi m\, k_{\text{B}}T_{\text{c}})^{3/2}}{\mathsf{h}^3} \\ \Rightarrow \quad \frac{N}{V} = n &= \frac{(2S+1)\zeta\left(\frac{3}{2}\right)}{\lambda_{\text{T}}^3(T_{\text{c}})} \\ \Rightarrow \quad n &\simeq \frac{2.612\,(2S+1)}{\lambda_{\text{T}}^3(T_{\text{c}})} \ , \end{aligned} \tag{9.25}$$

We may solve for T_{c}:

$$k_{\text{B}}T_{\text{c}} = \frac{\mathsf{h}^2}{2\pi m}\left(\frac{n}{2.612\,(2S+1)}\right)^{2/3} \ .$$

For temperatures $T < T_{\text{c}}$, there are a finite fraction of the particles residing in the ground state (i.e. $N_0 \propto N$) the *Bose-Einstein condensate* (BEC), with the remainder being in excited states, their number being Eq. (9.22). Note the fraction of particles residing in excited states,

[5] A less ad hoc justification is delineated in Appendix F (page 670) of the e-book Pathria, R.K. and P.D Beale, *Statistical Mechanics*, 3rd edition, Academic Press (2011)

$N_{\neq 0}$, tends to zero as $T \to 0$. Thus, rearranging Eq. (9.23),

$$\begin{aligned} N_0 &= N - N_{\neq 0} \\ &= N \left[1 - \left(\frac{T}{T_c} \right)^{3/2} \right] . \end{aligned} \tag{9.26}$$

Thus at $T = 0$, $N_0 = N$, while with increasing temperature the number in the ground state drops to zero (as a fraction of total number of bosons) at $T = T_c$.

The term *Bose-Einstein condensation* refers to the "condensation" of atoms into the ground state, not into a physical droplet of liquid. There are $N_0(T)$ bosons in the condensate and $N_{\neq 0}$ non-condensed bosons.

It is believed that the superfluid phase of ^{4}He below $2.7\,$K, is a BEC, but drastically modified because of the interactions between the atoms (it is not an ideal gas). However with the dilute ultracold alkali gases, one may measure the number of condensed atoms as a function of temperature.

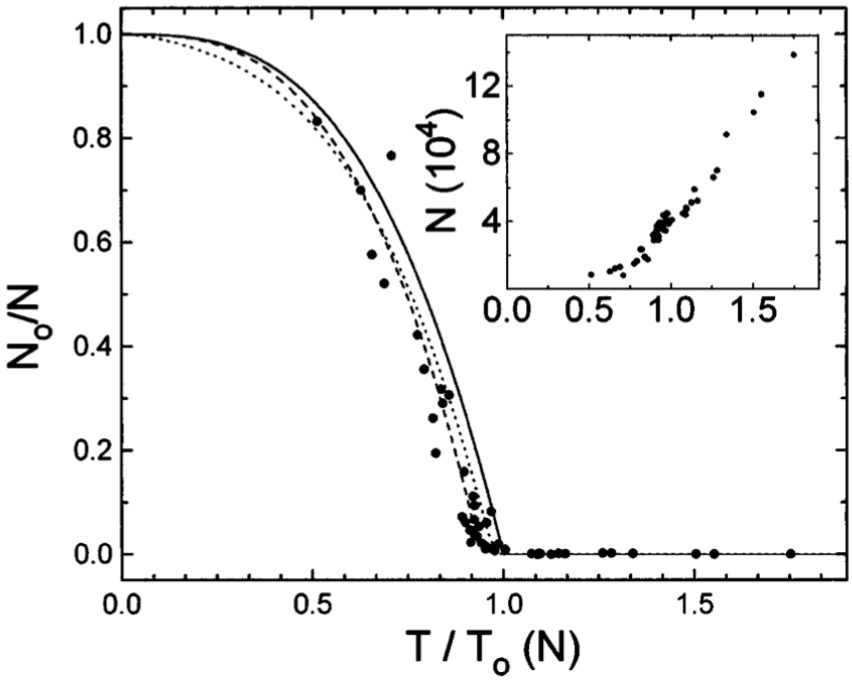

Figure 9.8: Total number N of ^{87}Rb atoms (inset) and ground-state fraction N_0/N as a function of scaled temperature T/T_0. The atoms are in a three-dimensional harmonic trap. The scale temperature $T_0(N)$ is the predicted critical temperature, in the thermodynamic (infinite N) limit, for an ideal gas in a harmonic potential. The solid (dotted) line shows the infinite (finite) N theory curves. At the transition, the cloud consists of 40 000 atoms at 280 nK. The dashed line is a least-squares fit to the form $N_0/N = 1 - (T/T_c)^3$ (note the exponent is 3 not 3/2 due to harmonic trap) which gives $T_c = 0.94(5)T_0$. Each point represents the average of three separate images. Credit: Ensher, J.R., D.S. Jin, M.R. Matthews, C.E. Wieman, and E.A. Cornell, *Physical Review Letters*, **77**, 4984 (1996).

9.9 Internal energy and pressure of the bose gas

We may directly write down $U = E^{\text{tot}}$ as:

$$U = \int_0^\infty \mathrm{d}\varepsilon\, \varepsilon\, g(\varepsilon)\, n^{\text{B}}(\varepsilon) = \frac{CV}{\beta^{5/2}} \int_0^\infty \frac{x^{3/2}\mathrm{d}x}{\mathrm{e}^x - 1} \simeq 1.783\, CV(k_{\text{B}}T)^{5/2} \ . \tag{9.27}$$

Where we have used (see Baierlein, *Thermal Physics*, CUP (1999), Appendix A, p422)

$$\int_0^\infty \mathrm{d}x\, \frac{x^{3/2}}{\mathrm{e}^x - 1} = \Gamma(5/2)\zeta(5/2) \simeq \frac{3}{4}\sqrt{\pi}\, 1.341 \simeq 1.783 \ .$$

Noting that the particles residing in the ground state, $\varepsilon = 0$, do not contribute to this result, we may exchange the dependence on V for a dependence on N, using Eq. (9.24), to obtain:

$$U \simeq 0.770\, k_{\text{B}}T\, N \left(\frac{T}{T_{\text{c}}}\right)^{3/2} \ . \tag{9.28}$$

The last two factors provide a factor of the number of bosons *not* in the ground state. The first two factors imply that each particle, with energy $\varepsilon > 0$, has an average energy of $0.77\, k_{\text{B}}T$. This should be compared to the average energy, $(3/2)\, k_{\text{B}}T$, it would have in the classical ideal monatomic gas.

9.10 The equation of state – for both bosons and fermions

In this section, for a general quantum system, we will write the First Law in terms of *changes* to the energies, E_i of each state i, and their probability of occupation $p_i = \exp(-\beta E_i)/Z$, as V and S are changed. At an intuitive level, as we will see, these reflect the work done and the change in entropy respectively. We will then restrict ourselves to ideal (quantum) gases and derive a surprisingly general expression for the equation of state.

Consider the expression for the energy written as a sum over *states* of the whole gas, labelled by i, with E_i being the states and p_i the probabilities:

$$p_i = \frac{\mathrm{e}^{-\beta E_i}}{Z} \Rightarrow E_i = -k_{\text{B}}T \ln(Z\, p_i) \tag{9.29}$$

Then

$$\begin{aligned} U = E &= \sum_i E_i p_i \\ \Rightarrow \quad \mathrm{d}U &= \sum_i (\mathrm{d}E_i\, p_i + E_i \mathrm{d}p_i) \\ \text{(Use Eq. (9.29).)} \quad &= \sum_i (p_i \mathrm{d}E_i - k_{\text{B}}T\, [\ln Z + \ln p_i]\, \mathrm{d}p_i) \ . \end{aligned} \tag{9.30}$$

Now note that

$$\sum_i p_i = 1 \Rightarrow \sum_i \mathrm{d}p_i = 0 \ . \tag{9.31}$$

And remembering the Shannon expression for entropy

$$\begin{aligned} S &= k_{\mathrm{B}} \sum_i p_i \ln p_i \\ \Rightarrow \quad \mathrm{d}S &= k_{\mathrm{B}} \sum_i \left(\mathrm{d}p_i \ \ln p_i + \frac{\mathrm{d}p_i}{p_i} p_i \right) \\ &= k_{\mathrm{B}} \sum_i \left(\mathrm{d}p_i \ \ln p_i + \underbrace{\mathrm{d}p_i}_{=0 \text{ upon summation}} \right) \\ \Rightarrow \quad \mathrm{d}S &= k_{\mathrm{B}} \sum_i \mathrm{d}p_i \ \ln p_i \ . \end{aligned} \tag{9.32}$$

So now combining the results Eq. (9.30), Eq. (9.31) and Eq. (9.32) we find

$$\mathrm{d}U = \sum_i p_i \, \mathrm{d}E_i - T \, \mathrm{d}S \ . \tag{9.33}$$

The E_i in general depend on the volume of the system – for example in one-dimension, the (non-degenerate) energy levels, n, of the infinite square well, of extent a, are

$$E_n = \frac{\hbar^2 (n\pi)^2}{2ma^2} \ .$$

The factor of a^{-2} is the dependence on the length of the well.

In general,

$$\begin{aligned} p_i \, \mathrm{d}E_i &= p_i \frac{\mathrm{d}E_i}{\mathrm{d}V} \mathrm{d}V \\ \Rightarrow \quad \sum_i p_i \, \mathrm{d}E_i &= \left\langle \frac{\mathrm{d}E}{\mathrm{d}V} \right\rangle \mathrm{d}V \ , \end{aligned} \tag{9.34}$$

where $\langle \cdots \rangle$ denotes an average with respect the probabilities $\{p_i\}$. Then inserting this result into Eq. (9.33), we see

$$\begin{aligned} \mathrm{d}U &= \left\langle \frac{\mathrm{d}E}{\mathrm{d}V} \right\rangle \mathrm{d}V - T \, \mathrm{d}S \\ &= -p \, \mathrm{d}V - T \, \mathrm{d}S \ . \end{aligned} \tag{9.35}$$

where we have identified the First Law of Thermodynamics, and hence

$$p = -\left\langle \frac{\mathrm{d}E}{\mathrm{d}V} \right\rangle \ . \tag{9.36}$$

We will now extract a simple result for the ideal Bose and Fermi gases from this general result. As usual we use periodic boundary conditions, so the allowed single particle momenta, $\mathbf{k}$, and

energies, $\varepsilon_{\mathbf{m}}$, are

$$\mathbf{k} = (k_x, k_y, k_z) = \frac{2\pi}{L}(m_x, m_y, m_z) = \frac{2\pi}{L}\mathbf{m}$$
$$\varepsilon_{\mathbf{m}} = \frac{(2\pi)^2}{L^2}\frac{\hbar^2(\mathbf{m})^2}{2m} = \frac{1}{V^{2/3}}\frac{\mathsf{h}^2(\mathbf{m})^2}{2m} \tag{9.37}$$

Then the energy, $E_{\{n_{\mathbf{m}}\}}$, of a state with $n_{\mathbf{m}}$ bosons (or fermions) in each of the states is

$$E_{\{n_{\mathbf{m}}\}} = \sum_{\mathbf{m}} \varepsilon_{\mathbf{m}} n_{\mathbf{m}} \tag{9.38}$$
$$= \frac{1}{V^{2/3}} \sum_{\mathbf{m}} \frac{\mathsf{h}^2(\mathbf{m})^2}{2m}\, n_{\mathbf{m}} \tag{9.39}$$

thus we may now make contact with the expression for the pressure, p, Eq. (9.36) using the label for the state, i, becoming the label for the many-boson (or many-fermion) state, $\{n_{\mathbf{m}}\}$, with the associated probability of that set of occupation numbers being $p_{\{n_{\mathbf{m}}\}}$, with the appropriate upper limit to the sums for bosons (infinity) and fermions (one),

$$U = E = \sum_i E_i\, p_i$$
$$= \sum_{\{n_{\mathbf{m}}\}} \sum_{\mathbf{m}} \frac{1}{V^{2/3}} \frac{\mathsf{h}^2(\mathbf{m})^2}{2m}\, n_{\mathbf{m}}\, p_{\{n_{\mathbf{m}}\}} \tag{9.40}$$
$$\Rightarrow \quad -\left\langle \frac{\mathrm{d}E}{\mathrm{d}V} \right\rangle = -\frac{\mathrm{d}}{\mathrm{d}V} \sum_{\{n_{\mathbf{m}}\}} \sum_{\mathbf{m}} \frac{1}{V^{2/3}} \frac{\mathsf{h}^2(\mathbf{m})^2}{2m}\, n_{\mathbf{m}}\, p_{\{n_{\mathbf{m}}\}}$$
$$= (-1)\left(-\frac{2}{3}\right) \frac{1}{V^{5/3}} \sum_{\{n_{\mathbf{m}}\}} \sum_{\mathbf{m}} \frac{\mathsf{h}^2(\mathbf{m})^2}{2m}\, n_{\mathbf{m}}\, p_{\{n_{\mathbf{m}}\}}$$
$$= \frac{2}{3}\frac{1}{V} \sum_{\{n_{\mathbf{m}}\}} \sum_{\mathbf{m}} \frac{1}{V^{2/3}} \frac{\mathsf{h}^2(\mathbf{m})^2}{2m}\, n_{\mathbf{m}}\, p_{\{n_{\mathbf{m}}\}}$$
(Use Eq. (9.40).) $$= \frac{2}{3}\frac{1}{V}\, U \ .$$

So in summary we have

$$p = -\left\langle \frac{\mathrm{d}E}{\mathrm{d}V} \right\rangle$$
$$= \frac{2}{3}\frac{1}{V}\, U$$
$$\Rightarrow \quad pV = \frac{2}{3} U \ . \tag{9.41}$$

This result, perhaps surprisingly, is *true for both bosons and fermions*. Earlier, we derived it for $T = 0$ for fermions.

In the fermionic case, we are hampered by not having an analytic expression for U for all T, but in the bosonic case we may substitute Eq. (9.28) into Eq. (9.41) to obtain:

$$pV \simeq k_{\mathrm{B}} T\, N \left\{ 0.520 \left(\frac{T}{T_{\mathrm{c}}}\right)^{3/2} \right\} \ . \tag{9.42}$$

The factor $\{\cdots\}$ modifies the classical ideal gas equation of state, reflecting that it is only the non-condensed particles which contribute to U.

9.11 Pressure, compressibility and entropy of a bose gas

We may write $U(T,V)$ by using Eq. (9.28) and Eq. (9.26), we see

$$U(T,V) \simeq 1.783 CV(k_{\text{B}}T)^{5/2} \ , \tag{9.43}$$

and so using Eq. (9.41) and Eq. (9.43) we obtain an expression for $p(T)$, :

$$p \simeq \tfrac{2}{3} C\,(k_{\text{B}}T)^{5/2} \ . \tag{9.44}$$

Note that P is *only* a function of T, below T_{c}, with no volume dependence! This is the same behaviour as the photon gas

Thus the bose gas is infinitely compressible as

$$\kappa_{\text{T}} = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T = 0 \ .$$

The entropy is given by

$$\begin{aligned} S &= \frac{U - \mu N - \Omega}{T} \\ \text{(But } \mu = 0.\text{)} \quad &= \frac{U - \Omega}{T} \\ &= \frac{5}{3}\frac{U}{T} \propto (k_{\text{B}}T)^{5/2} \end{aligned}$$

where we have used

$$\Omega = -pV = -\tfrac{2}{3}U \ .$$

The entropy resides in the non-condensed fraction, and hence drops to zero rapidly as $T \to 0$.

Finally since $G = \mu N$, the Gibbs free energy vanishes below the condensation temperature.